

TASVIRLARNI TAHLIL QILISH VA ANGLASH · Amaliy mashg'ulot (Practice)

16-mavzu. Gistogramma va nuqtaviy amallar

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Muhit. Python 3.11/3.12, faqat CPU. Versiyalar qat'iy belgilangan (to'liq ro'yxat — requirements.txt): numpy==2.4.4, opencv-python-headless==4.13.0.92, scikit-image==0.26.0, scipy==1.17.1, matplotlib==3.10.9. Barcha tasvirlar skimage.data ichida (camera, moon, coins, astronaut, brick) — internet kerak emas.

```
python -m venv .venv && source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
# har bir kodli topshiriq boshida:
import numpy as np, cv2
from skimage import data, exposure, filters, color
SEED = 42; rng = np.random.default_rng(SEED)
LOW = (90 + data.camera().astype(np.float64) * 70 / 255).round().astype(np.uint8) # past kontrastli camera
```

Mavzu. Nuqtaviy amal va LUT, gamma correction, contrast stretching, gistogramma, CDF, entropiya, histogram equalization, histogram matching (specification), CLAHE, Otsu ostonasi, gistogrammalarni taqqoslash, histogram backprojection, rangli tasvirlar.

Maqsadlar. Talaba: (1) LUT, CDF va tenglashtirish/moslash jadvarlarini qo'lda tuzadi; (2) asosiy teoremlarni (ehtimollik integral almashtirishi, $H(T(X)) \leq H(X)$) isbotlaydi; (3) OpenCV/scikit-image funksiyalarini o'z NumPy kodi bilan tekshiradi; (4) global usul, CLAHE va moslashni real tasvirlarda solishtirib, natijani sonlar bilan asoslaydi.

Baholash (100 ball). A bo'lim (1–13): $13 \times 2 = 26$; B bo'lim (14–17): $4 \times 3 = 12$; C bo'lim (18–30): $13 \times 4 = 52$; hisobot sifati (kod o'qiluvchanligi, chiqish formati, xulosalar): 10. Qo'lda hisoblashda oraliq qadamlar ko'rsatilmasa — yarim ball. Kod python -W error::DeprecationWarning bilan xatosiz ishlashi shart.

A. Qo'lda hisoblash (L — darajalar soni; yaxlitlash — eng yaqin butunga)

- Gamma va log.** $L = 256$, $r \in \{0, 50, 100, 150, 200, 255\}$ uchun negativ $s = 255 - r$, gamma correction $s = 255(r/255)^\gamma$ ($\gamma = 0,4$ va $\gamma = 2,5$) hamda $s = 255 \ln(1 + r) / \ln 256$ qiymatlarini jadvalga yozing.
- Bo'lakli chiziqli contrast stretching.** Tayanch nuqtalar $(r_1, s_1) = (70, 20)$, $(r_2, s_2) = (180, 230)$, $L = 256$. Uch bo'lak qiyaligini va $T(35)$, $T(120)$, $T(220)$ ni toping. Qaysi oraliqda kontrast kuchayadi?
- Min–max cho'zish.** 3 bitli ($L = 8$) 4×4 tasvir gistogrammasi $h = (0, 0, 5, 4, 4, 3, 0, 0)$. $[2, 5]$ oraliqni $[0, 7]$ ga cho'zuvchi LUT ni yozing; o'rtacha va standart chetlanishni oldin va keyin hisoblang.

- Gistogramma, CDF, entropiya.** 3 bitli tasvir $I = \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 3 \\ 2 & 3 & 3 & 7 \\ 1 & 2 & 6 & 7 \end{pmatrix}$ uchun $h(k)$, $p(k)$, $C(k)$, entropiya H

(bit), μ va σ ni toping.

- Histogram equalization (3 bit).** 64×64 , $L = 8$ tasvir: $h = (420, 980, 1210, 730, 390, 210, 106, 50)$. $s_k = \text{round}(7C(k))$ jadvalini tuzing, natijaviy gistogramma p_s va entropiyalarni (H_{oldin} , H_{keyin}) toping.
- OpenCV formulasi.** 5-topshiriq tasvirini $LUT(k) = \text{round}(7(\text{cdf}(k) - \text{cdf}_{\min}) / (N - \text{cdf}_{\min}))$ bo'yicha tenglashtiring. 5-topshiriq natijasidan qaysi daraja farq qiladi va nega?

- 4×4 tenglashtirish.** $I = \begin{pmatrix} 50 & 52 & 52 & 60 \\ 60 & 60 & 64 & 70 \\ 52 & 64 & 90 & 90 \\ 60 & 70 & 110 & 200 \end{pmatrix}$ ($L = 256$) uchun 6-topshiriqdagi formula bilan (7 o'rniga 255) LUT ni tuzing va natijaviy J matritsani yozing.

- Histogram matching.** 5-topshiriq tasvirini $p_z = (0,05; 0,10; 0,15; 0,20; 0,20; 0,15; 0,10; 0,05)$ ga moslang: $G(q) = 7C_z(q)$. (a) Yaxlitlangan s_k uchun $G(q)$ si eng yaqin q (teng masofada kichigi); (b) yaxlitlanmagan $7C(k)$ uchun $G(q) \geq 7C(k)$ bo'lgan eng kichik q . Ikkala LUT va (a) dagi natijaviy taqsimotni yozing.
- CLAHE kesish.** Katakda $M = 16$ piksel, $L = 8$, $h = (1, 0, 9, 4, 2, 0, 0, 0)$, ostonaga $\beta = 3$. Ortiqcha E , kesilgan $\tilde{h} = \min(h, \beta) + E/L$, $LUT = \text{round}(7 \text{cdf} / 16)$ ni toping va kesishsiz LUT bilan solishtiring.
- clipLimit.** 480×640 tasvir, `tileGridSize=(8,8)`, `clipLimit=3`. Katakda piksellar soni M va mutlaq ostonaga $\beta = \text{clipLimit} \cdot M / 256$ ni toping. Nega `clipLimit = 1` tekis gistogramma balandligiga teng?
- Otsu ostonasi.** $L = 8$, $h = (1, 4, 7, 3, 0, 2, 5, 2)$. $\sigma_B^2(t) = \omega_0 \omega_1 (\mu_1 - \mu_0)^2$ ni $t = 0, \dots, 6$ uchun hisoblang va t^* ni toping (teng qiymatda kichik t).

12. **Gistogrammalarni taqqoslash.** $H_1 = (0,4; 0,3; 0,2; 0,1)$, $H_2 = (0,1; 0,2; 0,3; 0,4)$ uchun Correlation, Chi-Square $\sum(H_1 - H_2)^2/H_1$, Intersection va Bhattacharyya $\sqrt{1 - \sum \sqrt{H_1 H_2}}$.
13. **LUT kompozitsiyasi.** T_1 — gamma $\gamma = 0,5$ (yaxlitlangan LUT), T_2 — negativ. $r \in \{0, 64, 128, 192, 255\}$ uchun $T_2 \circ T_1$ va $T_1 \circ T_2$ ni hisoblang. Kompozitsiya kommutativmi?

B. Isbot va konseptual savollar

14. **Uzluksiz holat.** $p_X(r) = 3r^2$, $r \in [0, 1]$. Tenglashtirish funksiyasi $T(r)$ ni toping va $T(X) \sim U[0, 1]$ ekanini isbotlang. So'ng X ni $p_Z(z) = 2(1 - z)$ taqsimotga moslovchi $T = G^{-1} \circ F_X$ ni oshkor yozing.
15. **Entropiya teoremasi.** Har qanday deterministik T uchun $H(T(X)) \leq H(X)$ ekanini isbotlang; tenglik shartini ayting. 5-topshiriq natijasi bilan izohlang.
16. **Qiyalik qoidasi.** $L = 256$. Gamma $\gamma = 0,4$ qaysi r larda kontrastni kuchaytiradi ($T'(r) > 1$)? $\gamma = 2,5$ uchun-chi? Chegaraviy r larni toping.
17. **Shovqin va rang.** (a) Tenglashtirish LUT qiyaligi $T'(k) \approx (L - 1)p(k)$ ekanini ko'rsating; $p(k) = 0,02$ bo'lganda shovqin necha marta kuchayadi? (b) Nega RGB kanallarini alohida tenglashtirish rang tusini buzadi, Lab L^* esa yo'q? (c) Nega CLAHE dan keyin entropiya oshishi mumkin, global LUT dan keyin esa yo'q?

C. Kodli topshiriqlar (har biri alohida funksiya; natijani ko'rsatilgan formatda chop eting)

18. **Qo'lda equalizeHist.** `equalize(img:np.ndarray)->np.ndarray` ni (`np.bincount`, `np.cumsum`, `cdf_min`) yozing. LOW uchun `cv2.equalizeHist` bilan maksimal farqni, hamda `camera`, LOW, natija uchun `std`, entropiya va noyob darajalar sonini chop eting. Format: `max|farq| = 0; asl std= 73.64 H=7.2317 daraja=256`.
19. **Statistika.** `hist_stats(img)->dict: mean, std, H, median, 2- va 98-persentil (CDF orqali np.searchsorted(cdf, q)), moda, p(0), p(255)`. `camera` va `moon` uchun bir qatordan chop eting; persentillarni `np.percentile` bilan solishtiring.
20. **Gamma LUT.** `gamma_lut(g)->uint8[256]`; `cv2.LUT(data.moon(), gamma_lut(g))` ni $g \in \{0,4; 0,7; 1; 1,5; 2,5\}$ uchun `exposure.adjust_gamma` bilan solishtiring. Jadval: `gamma mean std max|LUT-skimage|`.
21. **Persentil cho'zish.** `stretch(img, lo_pct=2, hi_pct=98)->(uint8, a, b)`. `data.moon()` uchun a, b , cho'zishdan oldin/keyin `std`, to'yingan (0 yoki 255) piksellar ulushi va `exposure.rescale_intensity(img, in_range=(a,b), out_range=(0,255))` bilan farqni chop eting.
22. **Histogram matching.** `match_hist(src, ref)->(out, lut): LUT = np.searchsorted(C_ref, C_src). src=LOW, ref=data.moon()`. O'rtacha va `std` (`src`, `ref`, qo'lda, `exposure.match_histograms`), LUT monotonligi, qo'lda va `skimage` orasidagi o'rtacha mutlaq farq, KS masofa $\max_k |C_{ref} - C_{out}|$ (4 xona).
23. **CLAHE:** `skimage.data.moon`. Asl, `cv2.equalizeHist` va `cv2.createCLAHE(clipLimit=c, tileGridSize=(8,8))`, $c \in \{1, 2, 4, 8\}$, hamda `exposure.equalize_adapthist(img, kernel_size=64, clip_limit=0.01)` (natija $\times 255$, yaxlitlab `uint8`) uchun `std` va entropiyani chop eting. `clipLimit` ning ta'sirini izohlang.
24. **Otsu.** `otsu(img)->(t, sigma_B2)` ni kumulyativ yig'indilar bilan $O(L)$ da yozing. `data.coins()` uchun `cv2.threshold(..., THRESH_OTSU)` va `filters.threshold_otsu` bilan solishtiring; $I > t$ piksellar ulushi (4 xona).
25. **Entropiya tajribasi.** LOW, `eq=cv2.equalizeHist(LOW)` va `eq(eq)` uchun entropiya va darajalar soni; `max|eq(eq)-eq|`. So'ng tasodifiy LUT `rng.integers(0,256,256) (seed=42)` ni qo'llab H oshmasligini tekshiring.
26. **Rangli tasvir.** `data.astronaut()`: (a) R, G, B alohida `equalizeHist`; (b) faqat Lab L^* kanali (`cv2.COLOR_RGB2LAB`). Har biri uchun o'rtacha doiraviy hue siljishini (`gradus, color.rgb2hsv`, ikkala tasvirda $S > 0,2$ piksellarda, $d = \min(|\Delta h|, 1 - |\Delta h|) \cdot 360$) va kulrang `std` ni chop eting.
27. **Notekis yoritish.** `img = round(data.brick() * np.linspace(0.25,1,512)[None,:])`. Asl, kirish, global `eq`, CLAHE (`clipLimit=2, 8 \times 8`) uchun chap va o'ng 128 ustunda o'rtacha yorqinlik va 15×15 oynadagi mahalliy `std` (`cv2.blur` orqali $\sqrt{E[x^2] - E[x]^2}$) o'rtachasini hamda ularning nisbatini jadval qilib chop eting. Qaysi usul nimani tuzatadi?
28. **compareHist.** 32 katakli, L1 normallangan gistogrammalar (`cv2.calcHist, cv2.normalize(..., norm_type=cv2.NORM_L1)`). `camera` ni o'zi, gamma 0,7 versiyasi, LOW va `moon` bilan 4 metrikada solishtiring (4 xonali jadval).
29. **Shovqin kuchayishi.** `noisy = clip(round(camera + rng.normal(0,3))) (seed=42)`. Osmon sohasi $[0:100, 0:150]$ da shovqin `std`: kirish (`noisy-camera`), `eq(noisy)-eq(camera)`, `medianBlur(3)` va `GaussianBlur((5,5),1.0)` dan keyin `eq`. Osmon modasi k uchun $255p(k)$ va LUT qiyaligini (`(lut[k+3]-lut[k-3])/6`) chop eting.

30. Histogram backprojection. `data.astronaut()` → HSV (`cv2.COLOR_RGB2HSV`); yuz namunasi `hsv[100:140, 200:240]`; H-S gistogramma `[30,32]` katak, `NORM_MINMAX` 0–255; `cv2.calcBackProject`; niqob `bp>50`. Chop eting: `bp` min/max/mean, niqob ulushi, niqob markazi (qator, ustun), yuz oynasi `[80:180,180:280]` dagi niqob ulushi.

Hujjatlar

- OpenCV Histograms: https://docs.opencv.org/4.x/d6/dc7/group__imgproc__hist.html
- OpenCV histogram equalization va CLAHE: https://docs.opencv.org/4.x/d5/daf/tutorial_py_histogram_equalization.html
- OpenCV backprojection: https://docs.opencv.org/4.x/dc/df6/tutorial_py_histogram_backprojection.html
- scikit-image exposure: <https://scikit-image.org/docs/stable/api/skimage.exposure.html>
- scikit-image filters (`threshold_otsu`): <https://scikit-image.org/docs/stable/api/skimage.filters.html>
- scikit-image data: <https://scikit-image.org/docs/stable/api/skimage.data.html>
- NumPy searchsorted: <https://numpy.org/doc/stable/reference/generated/numpy.searchsorted.html>